Mathematical Modeling

Chapter 5: Sensitivity Analysis and Lagrange Multipliers

Personal study notes

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1 What an optimal basis tells us

Suppose a linear program has already been solved. Sensitivity analysis asks what happens when its data change, and whether the final basis can be reused. We work with

$$\max c^T x, \qquad Ax = b, \qquad x \geq 0, \qquad A \in \mathbb{R}^{m \times n}, \quad \operatorname{rank} A = m.$$

An ordered basis $B = (j_1, \ldots, j_m)$ selects a nonsingular matrix A_B. Let N be the remaining indices, and define

$$\bar{A} = A_B^{-1} A, \quad \bar{b} = A_B^{-1} b, \quad y = A_B^{-T} c_B, \quad q = A^T y - c = \bar{A}^T c_B - c. \tag{1}$$

Our sign convention is $q_j = z_j - c_j$, the negative of the usual reduced cost. The basic solution and objective dictionary are

$$x_N = 0, \quad x_B = \bar{b}, \qquad c^T x = z_0 - q_N^T x_N, \quad z_0 = c_B^T \bar{b} = b^T y. \tag{2}$$

Thus $\bar{b} \geq 0$ establishes primal feasibility, and $q \geq 0$ establishes dual feasibility for $\min b^T y$ subject to $A^T y \geq c$, y free. Together they certify optimality.

Here an *optimal basis* means a basis passing both tests. An optimal point can have several bases, and at a degenerate optimal point a particular basis may fail $q \geq 0$. The calculations below follow the chosen certifying basis. Losing its certificate does not, by itself, show that an old degenerate point has lost optimality.

What is retained?	Meaning	What can still change?
Basis	Same column indices; both tests pass	Basic values, objective, dual prices
Point	Same decision vector remains optimal	Basis, objective value, multipliers
Optimal value	Same best objective value	Optimizers and bases

Keeping a basis is often the useful goal: it gives explicit formulas rather than requiring a new solve. Simultaneous changes are analyzable; they must be tested together, not against separate one-parameter intervals.

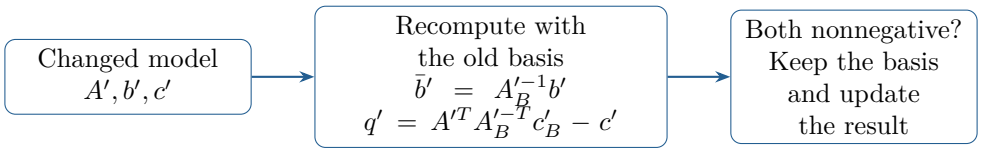


Figure 1: An optimal basis is a reusable certificate. The updated basis matrix must first be nonsingular.

2 Changing one right-hand side

Change only equation r:

$$b' = b + te_r, \qquad v = A_B^{-1} e_r, \qquad \bar{b}' = \bar{b} + tv.$$

Since A and c stay fixed, q and y stay fixed. The basis remains optimal exactly when every basic value stays nonnegative:

$$\max_{i: v_i > 0} \left(-\frac{\bar{b}_i}{v_i} \right) \leq t \leq \min_{i: v_i < 0} \left(-\frac{\bar{b}_i}{v_i} \right). \tag{3}$$

An empty maximum is $-\infty$, and an empty minimum is $+\infty$. A zero v_i imposes no new restriction because the starting $\bar{b}_i \geq 0$. Indices i refer to *basis rows*; r refers to an original equation.

Within this closed interval,

$$x_B(t) = \bar{b} + tv, \qquad z^*(b + te_r) = z_0 + tc_B^T v = z_0 + ty_r. \tag{4}$$

The multiplier y_r is the *shadow price*: objective change per unit of RHS change, while this basis is valid. At a finite endpoint a basic variable becomes zero. The same basis still certifies optimality there, but degeneracy and a different marginal value beyond the endpoint are possible.

Compute v by solving $A_B v = e_r$. A final tableau column gives v only if its original column is $+e_r$. A surplus column $-e_r$ gives $-v$. An artificial column is not part of the final original LP; retain the row transformation explicitly rather than relying on a deleted artificial.

2.1 A worked sensitivity tableau

Consider

$$\begin{aligned} \max \quad & -x_1 + 3x_2 - 2x_3 \\ \text{subject to} \quad & 3x_1 - x_2 + 2x_3 \leq 7, \\ & -2x_1 + 4x_2 \leq 12, \\ & -4x_1 + 3x_2 + 8x_3 \leq 10, \qquad x_1, x_2, x_3 \geq 0. \end{aligned} \tag{5}$$

Ratios are calculated in the column to the right of the RHS; $\longrightarrow$ in the separate Out column marks the chosen leaving row. The indicator row is labelled $z_j - c_j$; q_j abbreviates this difference in formulas. Vertical rules separate c_B, the basic-variable labels, the coefficient columns, and the RHS. Variable-column headings keep their order across pivots; only the basic-variable labels change. Artificial columns are removed after Phase I without reordering the retained columns.

Introduce slacks x_4, x_5, x_6. For $B = (1, 2, 6)$ the final tableau is

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	-1	3	-2	0	0	0	
-1	x_1	1	0	$4/5$	$2/5$	$1/10$	0	4
3	x_2	0	1	$2/5$	$1/5$	$3/10$	0	5
0	x_6	0	0	10	1	$-1/2$	1	11
	z_j	-1	3	$\frac{2}{5}$	$\frac{1}{5}$	$\frac{4}{5}$	0	11
	$z_j - c_j$	0	0	$12/5$	$1/5$	$4/5$	0	$z_0 = 11$

This gives $x^* = (4, 5, 0)$, slacks $(0, 0, 11)$, and $y = (1/5, 4/5, 0)$. The basis is nondegenerate and all nonbasic indicators are strictly positive, so the optimizer is unique.

Change the second RHS to $12 + t$. The x_5 column is v, hence

$$x_B(t) = \begin{pmatrix} 4 + t/10 \\ 5 + 3t/10 \\ 11 - t/2 \end{pmatrix}, \qquad \max\{-40, -50/3\} \leq t \leq 22.$$

Therefore

$$-\frac{50}{3} \leq t \leq 22, \qquad -\frac{14}{3} \leq b_2' \leq 34, \qquad z^* = 11 + \frac{4}{5}t. \tag{6}$$

The negative lower bound on b_2' is allowed: the row has a negative x_1 coefficient, so a negative RHS need not make it infeasible.

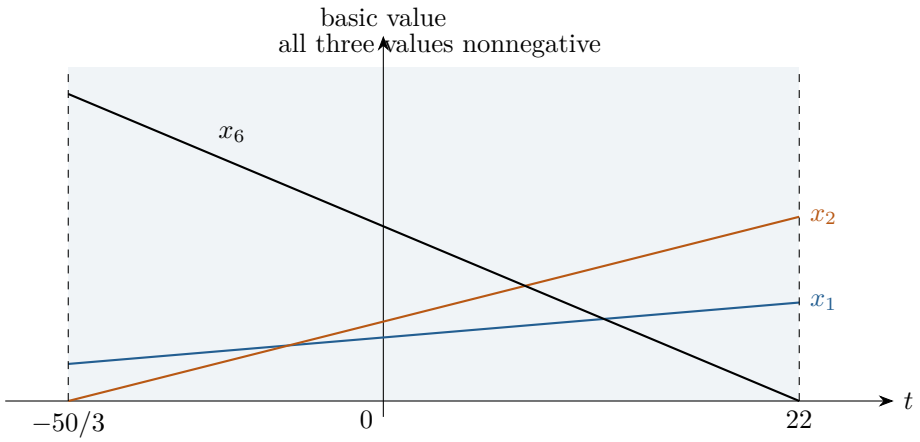


Figure 2: The allowable interval is the intersection of three basic-value inequalities. Its endpoints are determined by $x_2 = 0$ and $x_6 = 0$.

2.2 Endpoints, exercises, and the whole value function

b_2'	x_1	x_2	x_6	z^*
$-14/3$	$7/3$	0	$58/3$	$-7/3$
12	4	5	11	11
34	$31/5$	$58/5$	0	$143/5$

The endpoint calculation answers the exercise:

> Jakie jest RO zadania, gdy zmienimy $b_2 = -14/3$?

Here RO means an optimal solution. Substitute $t = -50/3$ in the basic vector, rather than substituting $-14/3$ for the *change*.

For the first RHS the x_4 column gives

$$(x_1, x_2, x_6) = (4 + 2t/5,\ 5 + t/5,\ 11 + t), \qquad t \geq -10, \quad b_1' \geq -3, \quad z^* = 11 + t/5.$$

Thus the answer to *Optymalna wartość f. celu ulega zmianie. O ile?* is $\Delta z = t/5$. The third RHS can change by $t \geq -11$: only $x_6 = 11 + t$ changes, and the value remains 11.

We can go beyond the basis interval for the second RHS, denoted now by r:

$$z^*(r) = \begin{cases} \text{infeasible}, & r < -14/3, \\ (4r + 7)/5, & -14/3 \leq r \leq 34, \\ 143/5, & r \geq 34. \end{cases} \tag{7}$$

To justify all three cases, first observe that x_3 can be set to zero: doing so relaxes constraints and increases the objective if $x_3 > 0$. The first constraint gives $x_1 \leq (7 + x_2)/3$, so $-2x_1 + 4x_2 \geq -14/3 + (10/3)x_2 \geq -14/3$. Below $-14/3$ feasibility is impossible. In the middle range, (6) supplies a feasible optimum and a dual certificate. For $r \geq 34$, $x = (31/5, 58/5, 0)$ is feasible, and $y = (9/5, 0, 8/5)$ is dual feasible with value $143/5$. Matching values prove the final branch.

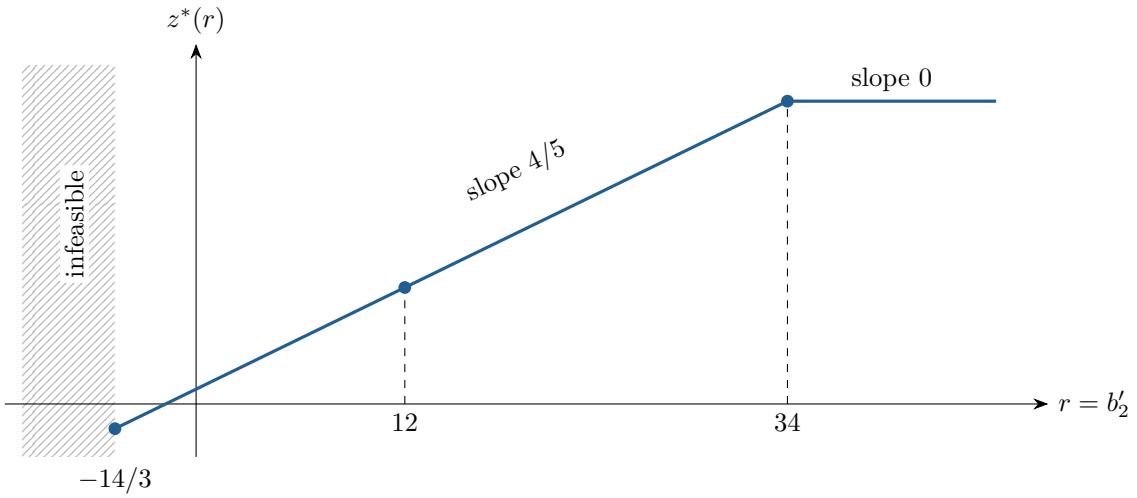


Figure 3: A shadow price applies to a branch of the value function. After $r = 34$, the second resource has zero marginal value.

3 Changing one objective coefficient

Changing c preserves every feasible point and the old basic vector. It changes the dual certificate and, usually, the value.

3.1 A nonbasic coefficient

If $k \in N$ and $c_k' = c_k + t$, then c_B and y are unchanged:

$$q_k' = q_k - t, \qquad q_j' = q_j \ (j \neq k).$$

The basis remains optimal for $t \leq q_k$. Its objective is unchanged because $x_k = 0$. For (5),

$$t \leq 12/5 \quad \Longleftrightarrow \quad c_3' \leq 2/5.$$

This answers *O ile może wzrosnąć wsp. c_3, nie wpływając na zmianę RO?* The maximum increase is 12/5, not the new coefficient 2/5.

At equality a zero indicator appears. To establish *distinct* optimizers, also check that a positive feasible step is possible. Here $x_3 = \theta$ gives

$$(x_1, x_2, x_6) = (4 - 4\theta/5,\ 5 - 2\theta/5,\ 11 - 10\theta), \qquad 0 \leq \theta \leq 11/10.$$

For $c_3' = 2/5$ this whole segment has value 11. For a larger c_3', even a small positive step improves the objective, so the old point is no longer optimal.

3.2 A basic coefficient

If $k = j_s \in B$ and $c_k' = c_k + t$, then $c_B' = c_B + te_s$ and

$$q_j' = q_j + t\bar{a}_{sj} \qquad (j \in N).$$

For basic columns the simultaneous change in c_k cancels, so $q_B' = 0$. The exact basis range is

$$\max_{j \in N : \bar{a}_{sj} > 0} \left(-\frac{q_j}{\bar{a}_{sj}} \right) \leq t \leq \min_{j \in N : \bar{a}_{sj} < 0} \left(-\frac{q_j}{\bar{a}_{sj}} \right). \tag{8}$$

Use original nonbasic columns only, not basic or removed artificial columns. A zero coefficient adds no restriction when $q_j \geq 0$. The old point remains fixed and $z' = z_0 + tx_k$.

For the main example:

Changed cost	From x_3	From x_4	From x_5	Allowable change	Value
$c_1 = -1 + t$	$t \geq -3$	$t \geq -1/2$	$t \geq -8$	$t \geq -1/2$	$11 + 4t$
$c_2 = 3 + t$	$t \geq -6$	$t \geq -1$	$t \geq -8/3$	$t \geq -1$	$11 + 5t$

Thus $c_1' \geq -3/2$ and $c_2' \geq 2$, with no upper bound. At $t = -1/2$ or -1 a nonbasic indicator vanishes and a positive step is available; alternative optimal points appear. Away from these endpoints, the original strict indicators give uniqueness.

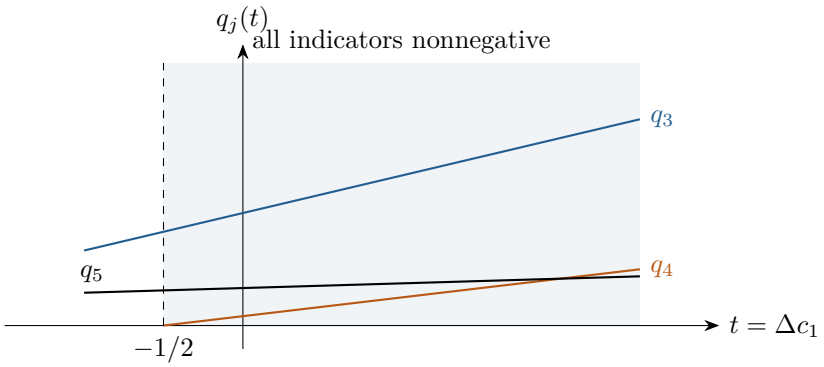


Figure 4: For a basic-cost change, several indicators move together. Here $q_4(t) = 1/5 + 2t/5$ determines the endpoint.

4 Vector changes and parameter regions

With fixed A, simultaneous changes $b' = b + \delta b$ and $c' = c + \delta c$ give

$$\bar{b}' = \bar{b} + A_B^{-1}\delta b, \qquad q' = q + \bar{A}^T \delta c_B - \delta c. \tag{9}$$

These are direct, coupled tests of the old basis. When both pass,

$$z' = (c_B + \delta c_B)^T A_B^{-1}(b + \delta b). \tag{10}$$

If only b changes, this reduces to $z_0 + y^T \delta b$. If only c changes, it reduces to $z_0 + x^T \delta c$. If both change, include the cross term $\delta c_B^T A_B^{-1} \delta b$.

One-at-a-time intervals cannot be combined independently. In (5), $\delta b = (0, 20, 0)$ gives $x_B = (6, 11, 1)$ and $\delta b = (0, 0, -10)$ gives $(4, 5, 1)$: both are allowable separately. Together they give $(6, 11, -9)$, which is infeasible. The exact joint region, rather than a rectangular collection of intervals, settles the question.

4.1 A smaller model for all subsequent updates

We will repeatedly reuse

$$\max\ 2x_1 - x_2 + x_3, \qquad x_1 + x_2 + x_3 \leq 6, \quad -x_1 + 2x_2 \leq 4, \quad x \geq 0. \tag{11}$$

Slacks are x_4, x_5, $B = (1, 5)$, and

$$A_B = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}, \qquad A_B^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

The inverse is displayed to understand the row transformation; numerical code should solve linear systems.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	RHS
	c_j	2	-1	1	0	0	
2	x_1	1	1	1	1	0	6
0	x_5	0	3	1	1	1	10
	z_j	2	2	2	2	0	12
	$z_j - c_j$	0	3	1	2	0	$z_0 = 12$

The unique decision optimum is $(6, 0, 0)$; $y = (2, 0)$.

For arbitrary new RHS (u, v) the same basis is optimal exactly when $u \geq 0$ and $u + v \geq 0$. The basic values are $(u, u + v)$ and the value is $2u$. For arbitrary decision costs $(\gamma_1, \gamma_2, \gamma_3)$ with slack costs zero, the basis is optimal exactly when

$$q_N = (\gamma_1 - \gamma_2,\ \gamma_1 - \gamma_3,\ \gamma_1) \geq 0.$$

This means $\gamma_1 \geq 0$, $\gamma_2 \leq \gamma_1$, and $\gamma_3 \leq \gamma_1$. These regions live in *data space*, not decision space.

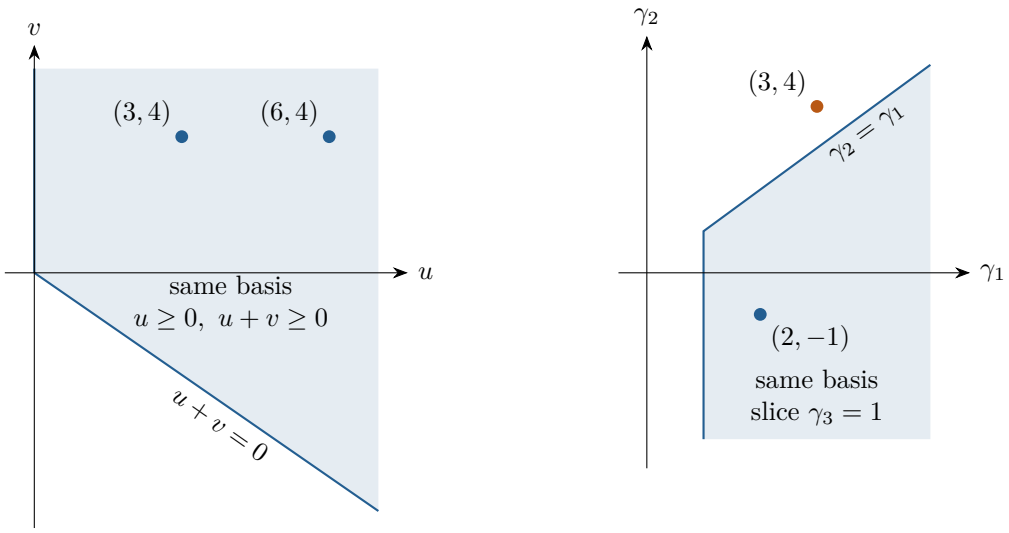


Figure 5: RHS feasibility and cost optimality regions for $B = (1, 5)$. The shaded regions continue beyond the plotting windows. The orange cost point is outside; the two marked RHS points are inside.

4.2 Reoptimize after changing the whole objective

Set decision costs to $(3, 4, 2)$. The old basic point is still feasible, its value is now 18, and its new indicator row is $(0, -1, 1, 3, 0)$.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	RHS	Ratio ($x_2 \uparrow$)	Out
	c_j	3	4	2	0	0			
3	x_1	1	1	1	1	0	6	$6/1 = 6$	
0	x_5	0	3	1	1	1	10	$10/3$	$\longrightarrow$
	z_j	3	3	3	3	0	18		
	$z_j - c_j$	0	-1	1	3	0	$z_0 = 18$		

Enter x_2 and remove x_5 with step $10/3$.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	RHS
	c_j	3	4	2	0	0	
3	x_1	1	0	2/3	2/3	−1/3	8/3
4	x_2	0	1	1/3	1/3	1/3	10/3
	z_j	3	4	$\frac{10}{3}$	$\frac{10}{3}$	$\frac{1}{3}$	$\frac{64}{3}$
	$z_j - c_j$	0	0	4/3	10/3	1/3	$z_0 = 64/3$

The new optimum is $(8/3, 10/3, 0)$, with dual $y = (10/3, 1/3)$. The gain from the *new starting value* is $10/3$, not $64/3 - 12$.

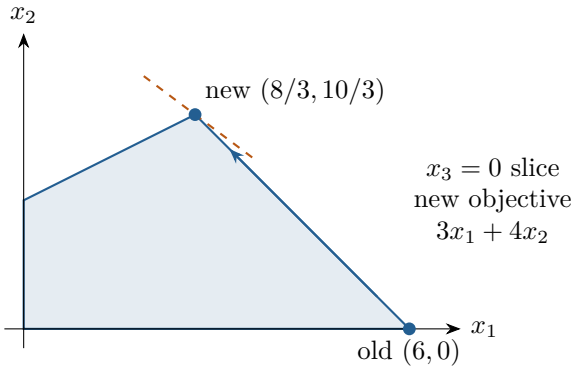


Figure 6: Changing the objective leaves the feasible set fixed and changes the supporting line. A single primal pivot follows the improving edge.

4.3 Change the RHS vector

Set $b' = (3, 4)$. The new basic vector is $(3, 7)$ and q is unchanged: $x = (3, 0, 0)$ is optimal, with value 6. No pivot is needed.

A failed basis test can also expose genuine infeasibility. For $b' = (6, -8)$, the basic values are $(6, -2)$ and the second canonical equation is

$$3x_2 + x_3 + x_4 + x_5 = -2.$$

Its left side is nonnegative. The contradiction certifies infeasibility; there is no eligible negative entry for a dual-simplex pivot.

5 Choosing a method after a model change

Change	What the old basis keeps	What to do if its test fails
Costs c	Primal feasibility	Recompute q; primal simplex
RHS b	Dual feasibility	Recompute $\bar{b}$; dual simplex
New nonnegative variable	Primal feasibility at new variable zero	Test its indicator; primal simplex
New $\leq$ constraint	Dual feasibility with a basic slack	Canonicalize the row; dual simplex
Both b and c	Neither guaranteed	Test both; choose a valid start
Matrix A	Neither guaranteed	Check nonsingularity, then both tests

This is a choice of starting method, not a guarantee that the modified problem has a finite optimum. Primal simplex can expose an improving ray; dual simplex can expose a contradictory row.

For a changed matrix A', the old column indices remain usable only if A'_B is nonsingular. Even one changed basic column can affect every transformed coefficient. If only a nonbasic column changes, $\bar{b}$ is unchanged and only that column's indicator needs retesting (when costs are unchanged). Use fresh linear solves rather than extending a one-parameter formula beyond its hypotheses.

6 Adding a nonnegative decision variable

Add $w \geq 0$ with column a and objective coefficient γ:

$$Ax + aw = b, \qquad \max(c^T x + \gamma w).$$

At $w = 0$ the old basic point is feasible. Append

$$u = A_B^{-1} a, \qquad q_w = c_B^T u - \gamma = y^T a - \gamma. \tag{12}$$

If $q_w \geq 0$, the old basis certifies optimality with $w = 0$. If $q_w < 0$, it loses its dual certificate; primal simplex can enter w. From a nondegenerate feasible basis, a positive feasible step improves the objective. For a degenerate basis the first step can be zero; reoptimization determines whether the old point is still optimal. If $u \leq 0$, the feasible direction $d_w = 1$, $d_B = -u$, $d_N = 0$ gives an unbounded improving ray.

For resource inequalities, $y^T a$ is the resource value of the new activity. Compare it with the activity's revenue γ. A negative component of a means the activity releases that resource, rather than consuming it.

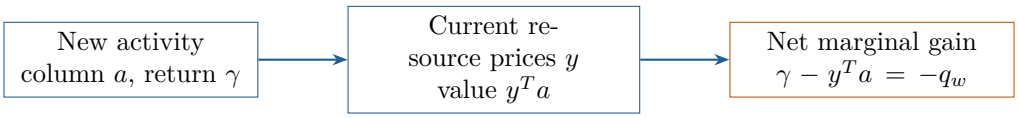


Figure 7: Pricing a new column gives an immediate test of the old dual certificate. Prices describe the current basis, not all later pivots.

6.1 Worked column update

For (11), introduce $x_6 \geq 0$ with

$$a = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \qquad \gamma = 1, \qquad u = \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \qquad q_6 = -3.$$

The canonical equations are now

$$x_1 + x_2 + x_3 + x_4 - x_6 = 6, \qquad 3x_2 + x_3 + x_4 + x_5 + x_6 = 10.$$

Enter x_6. Only the second row has a positive coefficient; the step is 10 and x_5 leaves. The final tableau is

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	2	-1	1	0	0	1	
2	x_1	1	4	2	2	1	0	16
1	x_6	0	3	1	1	1	1	10
	z_j	2	11	5	5	3	1	42
	$z_j - c_j$	0	12	4	5	3	0	$z_0 = 42$

The extended optimum is $(16, 0, 0, 0, 0, 10)$, with objective $2(16) + 10 = 42$. The gain is $(-q_6)10 = 30$. Both original inequalities with the new column are tight, and $y = (5, 3)$ is dual feasible with value $6(5) + 4(3) = 42$.

7 Adding a constraint

A new constraint shrinks the feasible set. If an old optimizer satisfies it, that point remains optimal: no point in the smaller set can beat the old maximum. If the old point violates it, a new feasible point is required.

For a new inequality $a^T x \le \beta$, introduce a slack $s \ge 0$: $a^T x + s = \beta$. Substitute $x_B = \bar{b} - \bar{A}_N x_N$ to obtain

$$s + (a_N^T - a_B^T \bar{A}_N) x_N = \beta - a_B^T \bar{b}. \tag{13}$$

The old basic columns have been eliminated from this row. With s basic and zero cost, the enlarged basis is nonsingular and its indicator row is $(q, 0)$. The new slack value is the constraint residual at the old point. A negative residual gives a dual feasible, primal infeasible start for dual simplex. Merely appending the untransformed row does not produce a canonical tableau.

For a new equality, no permanent slack can absorb a residual. If the old point violates it, one option is a signed artificial: $a^T x + \sigma w = \beta$, $w \ge 0$, where σ is chosen so that $w = |\beta - a^T x^*|$ at the old point. Phase I minimizes w. A positive Phase I minimum certifies infeasibility. At zero, remove the artificial and restore the true objective; a zero basic artificial needs pivoting out or redundant-row cleanup. Equalities already satisfied by the old point need no change of optimizer, though a basis representation may need adjustment.

7.1 Worked row update and a dual pivot

Add $x_1 - 2x_3 \le -2$ to (11). The old point $(6, 0, 0)$ violates the row. With new slack x_6,

$$x_1 - 2x_3 + x_6 = -2.$$

Subtract the old x_1 row. The new canonical tableau is

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Dual ratios ($r = 3$)	Out
	c_j	2	-1	1	0	0	0			
2	x_1	1	1	1	1	0	0	6	$x_2:\ 3/(1) = 3$	
0	x_5	0	3	1	1	1	0	10	$x_3:\ 1/(3) = 1/3$	
0	x_6	0	-1	-3	-1	0	1	-8	$x_4:\ 2/(1) = 2$	$\longrightarrow$
	z_j	2	2	2	2	0	0	12		
	$z_j - c_j$	0	3	1	2	0	0	$z_0 = 12$		

Select the negative RHS row and compare the dual ratios displayed to the right. Enter x_3, leave x_6, and pivot on -3:

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	2	-1	1	0	0	0	
2	x_1	1	2/3	0	2/3	0	1/3	10/3
0	x_5	0	8/3	0	2/3	1	1/3	22/3
1	x_3	0	1/3	1	1/3	0	$-1/3$	8/3
	z_j	2	$\frac{5}{3}$	1	$\frac{5}{3}$	0	$\frac{1}{3}$	$\frac{28}{3}$
	$z_j - c_j$	0	8/3	0	5/3	0	1/3	$z_0 = 28/3$

The optimum is $(10/3, 0, 8/3)$ with value $28/3$. Its slacks are $(0, 22/3, 0)$; $y = (5/3, 0, 1/3)$ gives the matching dual value $6(5/3) + 4(0) - 2(1/3) = 28/3$. The value decreases from 12, as a smaller feasible set requires.

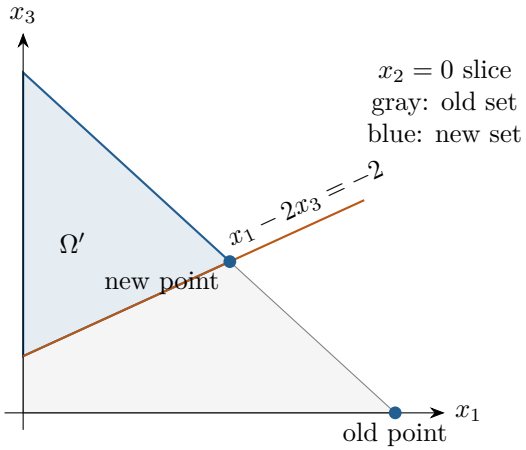


Figure 8: The new inequality cuts off the old optimizer. This is the $x_2 = 0$ slice; the tableau certifies the optimum in the full space.

8 Removing a variable

Removing x_s means imposing $x_s = 0$ and deleting its column and cost. The feasible set, embedded in the old space, becomes smaller. If an old optimizer has $x_s = 0$, deleting that coordinate gives an optimizer of the reduced problem. A nonbasic zero column can simply be removed. A *basic* zero variable requires a basis exchange or redundant-row cleanup before its column is removed.

If $x_s^* > 0$, feasibility of the reduced model is not assured. A reliable feasibility phase is

$$\min x_s, \qquad Ax = b, \quad x \geq 0.$$

The old point provides a feasible start. A strictly positive minimum proves that setting $x_s = 0$ is impossible. At a zero minimum, remove the column after cleaning up the basis, then restore and optimize the reduced objective. A zero-valued variable can still be basic: basis membership alone is not an infeasibility certificate. An unqualified finite Big M is not a substitute for this feasibility test.

Because the original maximum is finite, any feasible reduced model is bounded above by it. For a nonempty polyhedron, a bounded linear objective attains its maximum. Here the possible outcomes are therefore finite optimum or infeasibility, rather than a newly unbounded maximum.

Deletion	Feasible-set effect	Value effect for maximization
Variable	Impose its value zero	Cannot increase; may become infeasible
Constraint	Relax the restrictions	Cannot decrease; may become unbounded

For (11), deleting x_2 retains $(6, 0, 0)$ and value 12. Deleting x_1 gives

$$\max(-x_2 + x_3), \qquad x_2 + x_3 \leq 6, \quad 2x_2 \leq 4, \quad x_2, x_3 \geq 0.$$

Since $-x_2 + x_3 \leq x_2 + x_3 \leq 6$, $(x_2, x_3) = (0, 6)$ is optimal. By contrast, in $x_1 + x_2 = 1$, $x_2 = 0$, $x \geq 0$, deletion of x_1 leaves the impossible equation $0 = 1$.

9 Removing a constraint

Removing a constraint enlarges the feasible set. The old point stays feasible, but may cease to be optimal; a finite old optimum can become unbounded.

9.1 When the old dual certificate survives

For $\max c^T x$ subject to $Gx \le b$, $x \ge 0$, a dual certificate is $y \ge 0$, $G^T y \ge c$. If some optimal dual vector has $y_r = 0$, drop its rth component. The remaining vector is dual feasible for the model without row r, and its objective is unchanged. The old primal point still has the same value, so it remains optimal.

Strict slack at an old optimal point implies $y_r = 0$ by complementarity and is sufficient for this conclusion. It is not necessary: a tight row can also have zero multiplier. For equalities the multipliers are free; a zero multiplier similarly permits deletion of that row. Checking one chosen dual vector is a sufficient test; failure of that test alone does not determine the best new value.

9.2 Relaxing a row without throwing away the tableau

Deleting equation r can be represented by

$$Ax + e_r u - e_r v = b, \qquad u, v \ge 0, \quad c_u = c_v = 0.$$

The difference $u - v$ can absorb any residual of the deleted equation. The other equations are unchanged. Original feasible vectors of the reduced model correspond exactly to extended feasible vectors after choosing u, v. The new indicators are $q_u = y_r$ and $q_v = -y_r$. If either is negative, primal simplex can improve the old solution.

For a converted $\le$ row, its existing slack already has column $+e_r$. Add only the opposite column $-e_r$ with zero cost. For a converted $\ge$ row, the existing surplus is $-e_r$, so add $+e_r$. Once solved, discard the auxiliary pair and the deleted equation. This construction can also expose unboundedness.

9.3 Production model: two different deletions

Consider the production model

$$\max\ 3x_1 + 2x_2, \quad 2x_1 + x_2 \le 10, \quad 3x_1 + 3x_2 \le 24, \quad \tfrac{3}{2}x_1 \le 6, \quad x \ge 0. \tag{14}$$

The optimum is $(2, 6)$, value 18, with slacks $(x_3, x_4, x_5) = (0, 0, 3)$ and dual prices $(1, 1/3, 0)$.

Removing the third constraint retains the optimum: its slack is positive, its dual multiplier is zero, and the first two rows still bound the objective by $3x_1 + 2x_2 = (2x_1 + x_2) + (x_1 + x_2) \le 10 + 8 = 18$.

Removing the first constraint changes the result. The remaining rows imply $x_1 \le 4$ and $x_1 + x_2 \le 8$, giving

$$3x_1 + 2x_2 = x_1 + 2(x_1 + x_2) \le 4 + 16 = 20.$$

The point $(4, 4)$ attains 20. This is an exact bound, not just a graphical guess.

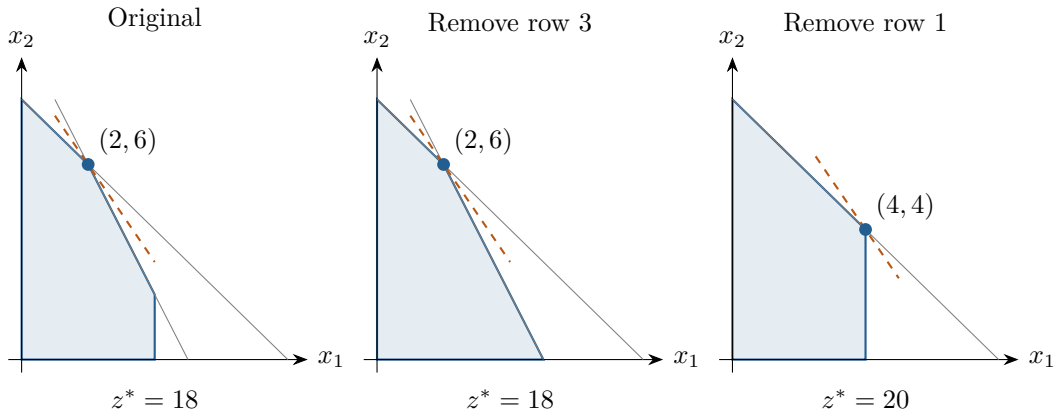


Figure 9: Constraint-removal geometry: deleting a slack row preserves the optimizer; deleting a priced row permits improvement. All three sets enlarge or stay fixed relative to the relevant retained inequalities.

Here is the full update for deleting row 1 using x_6 with column $-e_1$. Start from the ordered basis $B = (2, 5, 1)$:

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS	Ratio ($x_6 \uparrow$)	Out
	c_j	3	2	0	0	0	0			
2	x_2	0	1	-1	$2/3$	0	1	6	$6/1 = 6$	
0	x_5	0	0	$-3/2$	$1/2$	1	$3/2$	3	$3/(3/2) = 2$	$\longrightarrow$
3	x_1	1	0	1	$-1/3$	0	-1	2	–	
	z_j	3	2	1	$\frac{1}{3}$	0	-1	18		
	$z_j - c_j$	0	0	1	$1/3$	0	-1	$z_0 = 18$		

Enter x_6 and remove x_5 with step 2.

c_B	Basic	x_1	x_2	x_3	x_4	x_5	x_6	RHS
	c_j	3	2	0	0	0	0	
2	x_2	0	1	0	$1/3$	$-2/3$	0	4
0	x_6	0	0	-1	$1/3$	$2/3$	1	2
3	x_1	1	0	0	0	$2/3$	0	4
	z_j	3	2	0	$\frac{2}{3}$	$\frac{2}{3}$	0	20
	$z_j - c_j$	0	0	0	$2/3$	$2/3$	0	$z_0 = 20$

The auxiliary $x_6 = 2$ is precisely the violation $2(4) + 4 - 10 = 2$ of the deleted inequality. Discarding x_3, x_6 gives the reduced solution $(x_1, x_2, x_4, x_5) = (4, 4, 0, 0)$. The remaining-row multipliers $(2/3, 2/3)$ have value $24(2/3) + 6(2/3) = 20$.

Deleting a row can also cause unboundedness: in $\max x_1$ subject to $x_1 \le 1$, $x_1 \ge 0$, removal of $x_1 \le 1$ leaves the improving ray $x_1 = \theta$, $\theta \ge 0$.

10 Lagrange multipliers and active constraints

Write all inequalities in the same orientation:

$$\max c^T x, \qquad g_i(x) = a_i^T x - b_i \le 0 \ (i \in I), \qquad h_j(x) = a_j^T x - b_j = 0 \ (j \in E).$$

Here $x \in \mathbb{R}^n$ is otherwise unrestricted. Include variable bounds explicitly among the inequalities: $x_k \ge 0$ means $g(x) = -x_k \le 0$. An inequality is *active* at $\hat{x}$ if $g_i(\hat{x}) = 0$. Denote these indices by $\mathcal{A}(\hat{x})$.

10.1 Feasible directions and outward normals

For a feasible point $\hat{x}$, a direction d allows sufficiently small positive steps $\hat{x} + \alpha d$ exactly when

$$a_i^T d \le 0 \quad (i \in \mathcal{A}(\hat{x})), \qquad a_j^T d = 0 \quad (j \in E).$$

Inactive rows impose no restriction on a sufficiently small step: each has a strictly positive slack. Because there are finitely many rows, one positive step bound can be chosen for all of them. Optimality is equivalent to $c^T d \le 0$ for every such direction.

If the objective normal can be represented as

$$c = \sum_{i \in I} \lambda_i a_i + \sum_{j \in E} \nu_j a_j, \qquad \lambda_i \ge 0, \quad \lambda_i = 0 \text{ for } i \notin \mathcal{A}(\hat{x}), \tag{15}$$

then its product with every feasible direction is nonpositive. Equality normals can have coefficients of either sign; inequality normals must point outward with nonnegative weights.

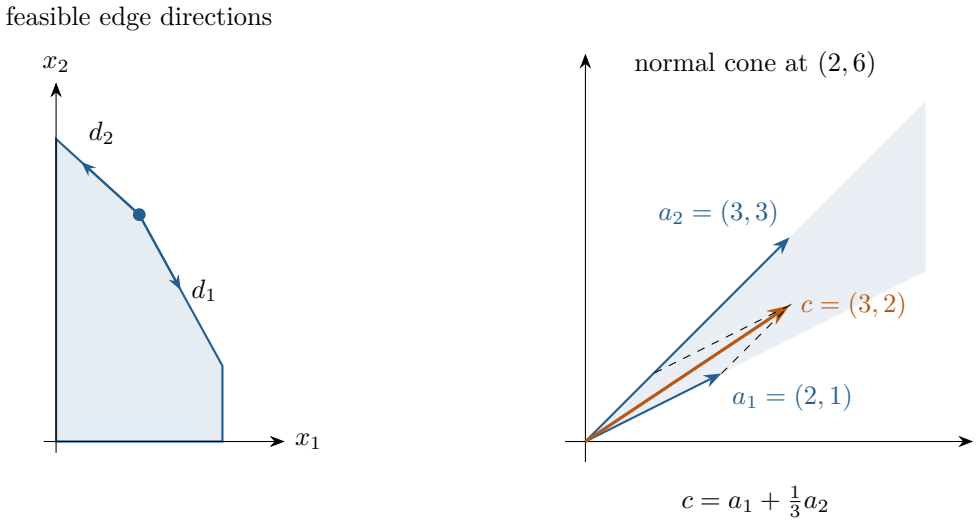


Figure 10: For the production optimum, c lies between the active outward normals. Its products with both feasible edge directions are negative. The right panel is a space of normals, not another feasible region.

10.2 The Lagrangian and the four KKT conditions

It is convenient to minimize the negative objective:

$$L(x, \lambda, \nu) = -c^T x + \sum_{i \in I} \lambda_i g_i(x) + \sum_{j \in E} \nu_j h_j(x). \tag{16}$$

Then the Karush–Kuhn–Tucker (KKT) conditions are

Condition	Formula
Primal feasibility	$g_i(\hat{x}) \le 0, \quad h_j(\hat{x}) = 0$
Multiplier signs	$\lambda_i \ge 0, \quad \nu_j \in \mathbb{R}$
Complementarity	$\lambda_i g_i(\hat{x}) = 0$
Stationarity	$-c + \sum_i \lambda_i a_i + \sum_j \nu_j a_j = 0$

For this linear program, these conditions are necessary and sufficient for a feasible point to be optimal.

For sufficiency, compare any feasible x with $\hat{x}$:

$$c^T(x-\hat{x}) = \sum_i \lambda_i a_i^T(x-\hat{x}) + \sum_j \nu_j a_j^T(x-\hat{x}) \le 0.$$

Equalities contribute zero; positive inequality multipliers have active rows, so their contributions are nonpositive. For necessity, apply linear-programming strong duality to an optimal feasible point, including its bound inequalities. An optimal dual vector exists. The primal–dual gap is a sum of nonnegative multiplier–slack products; zero gap gives complementarity, and the dual equations give stationarity. No independence of active inequality normals is required.

For a nonlinear problem, replace c and a_i by the appropriate gradients. For a differentiable concave maximization objective with convex inequalities and affine equalities, KKT conditions still suffice globally. Necessity generally requires a constraint qualification, such as Slater's condition (with the usual domain assumptions); convexity alone is insufficient. For example, $\max x$ subject to $x^2 \le 0$ has the unique optimum $x = 0$, but stationarity would demand $-1+\lambda(2x) = 0$, which is impossible there.

10.3 A complete mixed-sign multiplier certificate

Consider the question:

Czy $\hat{x} = [0,\ 17/4,\ 0,\ 23/4]^T$ jest RO?

The model is

$$\begin{aligned} \max \quad & x_1 + 3x_2 - 2x_3 + x_4 \\ \text{subject to} \quad & 3x_1 + 2x_2 + x_3 + x_4 \le 20, \\ & 2x_1 + x_2 + x_4 = 10, \\ & 5x_1 - 2x_2 + x_3 + 2x_4 \ge 3, \qquad x \ge 0. \end{aligned} \tag{17}$$

Orient the third row as $3 - 5x_1 + 2x_2 - x_3 - 2x_4 \le 0$. Assign λ_1 to row 1, ν to the equality, λ_3 to the oriented row 3, and μ_k to the bound $-x_k \le 0$. At $\hat{x}$ the row values are 57/4, 10, and 3.

Constraint	Nonnegative slack	Active?	Multiplier
Row 1 (≤ 20)	23/4	No	$\lambda_1 = 0$
Equality row	0	Yes	$\nu = 2$ (free)
Row 3 (≥ 3)	0	Yes	$\lambda_3 = 1/2$
$x_1 \ge 0$	0	Yes	$\mu_1 = 1/2$
$x_2 \ge 0$	17/4	No	$\mu_2 = 0$
$x_3 \ge 0$	0	Yes	$\mu_3 = 3/2$
$x_4 \ge 0$	23/4	No	$\mu_4 = 0$

Inactive-row multipliers vanish by complementarity. Stationarity in coordinates 2 and 4 gives

$$-3 + \nu + 2\lambda_3 = 0, \qquad -1 + \nu - 2\lambda_3 = 0.$$

Hence $\nu = 2$ and $\lambda_3 = 1/2$. The remaining coordinates give

$$-1 + 2\nu - 5\lambda_3 - \mu_1 = 0, \qquad 2 - \lambda_3 - \mu_3 = 0,$$

so $\mu_1 = 1/2$ and $\mu_3 = 3/2$. All inequality multipliers are nonnegative. Equivalently, the full normal identity is

$$\begin{pmatrix} 1 \\ 3 \\ -2 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 1 \\ 0 \\ 1 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} -5 \\ 2 \\ -1 \\ -2 \end{pmatrix} - \frac{1}{2} e_1 - \frac{3}{2} e_3.$$

For any feasible x this yields the explicit upper bound

$$c^T x = 2(2x_1 + x_2 + x_4) + \frac{1}{2}(-5x_1 + 2x_2 - x_3 - 2x_4) - \frac{1}{2}x_1 - \frac{3}{2}x_3 \le 20 - \frac{3}{2} = \frac{37}{2}.$$

The candidate attains $37/2$, so it is optimal. It is also unique: equality in this bound forces $x_1 = x_3 = 0$ and the third row tight. Together with $x_2 + x_4 = 10$ these give $x_2 = 17/4$, $x_4 = 23/4$.

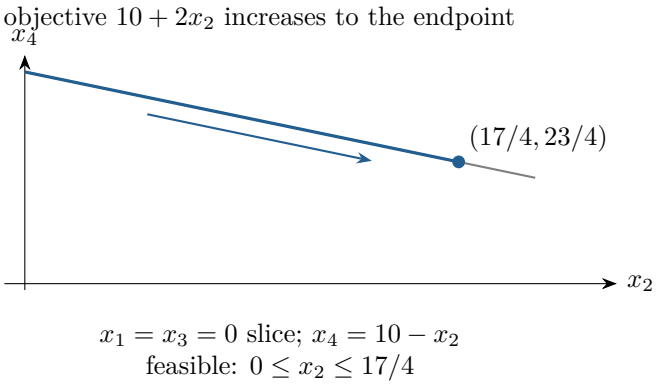


Figure 11: The two zero bound coordinates reveal a one-dimensional slice. The multiplier certificate proves optimality in all four dimensions.

11 Connecting multipliers, prices, and reduced costs

For $\max c^T x$ with $Gx \le b$, $x \ge 0$, include bounds $-x \le 0$ in the Lagrangian. Stationarity reads

$$-c + G^T y - \mu = 0, \qquad y \ge 0, \quad \mu \ge 0.$$

Thus $\mu = G^T y - c$ is the vector of decision-variable indicators. Complementarity is simultaneously $y_i(b_i - G_{i:}x) = 0$ and $\mu_j x_j = 0$. For standard equalities, use free y and the same $\mu = A^T y - c = q$. These are the same optimality conditions expressed through a tableau, a dual problem, and a Lagrangian.

Quantity	Algebra	Interpretation
RHS multiplier	$y = A_B^{-T} c_B$	Local value of relaxing an RHS
Decision indicator	$q_j = a_j^T y - c_j$	Marginal loss from a nonbasic activity
Bound multiplier	$\mu_j = q_j$	Nonnegative weight on $-x_j \le 0$
Active-row multiplier	$\lambda_i \ge 0$	Weight of its outward normal

A tight row need not have a positive price. Prices can be nonunique, especially at degeneracy. For fixed A, c, the maximum value $V(b) = \min_{A^T y \ge c} b^T y$ is concave in b on the region of finite feasible optima: it is a pointwise infimum of affine functions. An old optimal y gives the global upper bound

$$V(b + \delta b) \le V(b) + y^T \delta b$$

whenever the changed primal is feasible with a finite optimum. Equality holds when the same certifying basis remains primal feasible. At a kink, the bound remains valid even though one global derivative need not exist. For fixed feasible set, the maximum value as a function of c is convex, because it is a supremum of linear functions $c^T x$. These curvature statements use *maximization*; minimization reverses them.

12 Beyond LP: what nonlinear KKT conditions establish

For differentiable minimization with $x \in \mathbb{R}^n$,

$$\min f(x), \qquad g_i(x) \le 0, \qquad h_j(x) = 0, \qquad L = f + \sum_i \lambda_i g_i + \sum_j \nu_j h_j,$$

stationarity becomes $\nabla f + \sum_i \lambda_i \nabla g_i + \sum_j \nu_j \nabla h_j = 0$. Feasibility, $\lambda \ge 0$ and $\lambda_i g_i = 0$ are still required. They must be interpreted with the following distinctions.

- If f, g_i are convex and h_j are affine, every KKT point is a global minimizer. For any feasible x, complementarity gives $L(\hat{x}, \lambda, \nu) = f(\hat{x})$ and the signs give $L(x, \lambda, \nu) \le f(x)$. Convexity and stationarity make $\hat{x}$ minimize L, hence $f(\hat{x}) \le f(x)$.
- In this differentiable convex setting with functions defined on $\mathbb{R}^n$, a sufficient Slater condition is a point satisfying all equalities and $g_i(x) < 0$ for every inequality. If a minimizer exists, this condition guarantees KKT multipliers. More general domains use relative-interior qualifications; strict feasibility is sufficient, not necessary for every model, particularly LPs.
- In a general smooth problem, a local minimizer admits KKT multipliers under suitable regularity. For example, linear independence of the active inequality gradients together with the equality gradients is the *linear independence constraint qualification* (LICQ). Even with this qualification, a KKT point need not be a minimum.

For a direct counterexample, minimize $f(t) = t^3$ on $[-1, 1]$. At $t = 0$ both bound inequalities are inactive, their multipliers are zero, and $f'(0) = 0$: all KKT conditions hold. Yet negative nearby t have smaller value, so this point is not even a local minimum. The failure is nonconvexity, not a missing multiplier. Likewise, $f(t) = -t^2$ has a stationary local maximum at zero. First-order stationarity alone does not classify a point.

If functions are twice differentiable, second-order analysis examines $\nabla^2_{xx} L$ on appropriate feasible or critical directions, not necessarily on all of $\mathbb{R}^n$. For instance, minimize $u^2 - v^2$ subject to $v = 0$. At $(0, 0)$ the Hessian is $\mathrm{diag}(2, -2)$, but every nonzero tangent direction is $(d, 0)$ and has curvature $2d^2 > 0$; the constrained point is the unique global minimum. For unconstrained problems, a positive-definite Hessian at a stationary point suffices for a strict local minimum; a positive-semidefinite Hessian alone does not, as t^3 shows.

13 A first look at barriers and the central path

Complementarity also explains the link to interior-point methods. For the standard maximization LP, minimize $-c^T x$ and introduce a logarithmic barrier with parameter $\tau > 0$:

$$\min_{Ax=b,\ x>0} \left(-c^T x - \tau \sum_{j=1}^{n} \log x_j \right).$$

Here $x > 0$ means every coordinate is strictly positive; assume such a feasible point exists and that the barrier subproblem has a minimizer. The logarithm acts on nondimensionalized coordinates. With positive reference scales, $\log(x_j / s_j)$ gives the same minimizers by adding a constant. The barrier tends to $+\infty$ as a coordinate approaches zero, keeping the subproblem's solutions in the interior relative to $Ax = b$.

Using a free equality multiplier y, stationarity gives

$$-c + A^T y - \tau \begin{pmatrix} 1/x_1 \\ \vdots \\ 1/x_n \end{pmatrix} = 0.$$

Set $q = A^T y - c$. Then a barrier optimum satisfies

$$Ax = b, \quad x > 0, \qquad q > 0, \qquad x_j q_j = \tau \quad (j = 1, \ldots, n). \tag{18}$$

Both primal and dual are feasible and their gap is $b^T y - c^T x = x^T q = n\tau$. Decreasing τ therefore approaches zero complementarity gap. The family of exact subproblem solutions is the *central path*; actual algorithmic iterates need not lie exactly on it.

Consider $\max x_1$ subject to $x_1 + x_2 = 1$, $x \geq 0$. Here $y = \tau/x_2$ and $y - 1 = \tau/x_1$; eliminating y gives

$$x_1(\tau) = \frac{1 - 2\tau + \sqrt{1 + 4\tau^2}}{2}, \qquad x_2 = 1 - x_1.$$

For every $\tau > 0$, $1/2 < x_1 < 1$; as $\tau \downarrow 0$, x approaches the boundary optimum $(1, 0)$. The dual slack pair stays positive until the limit.

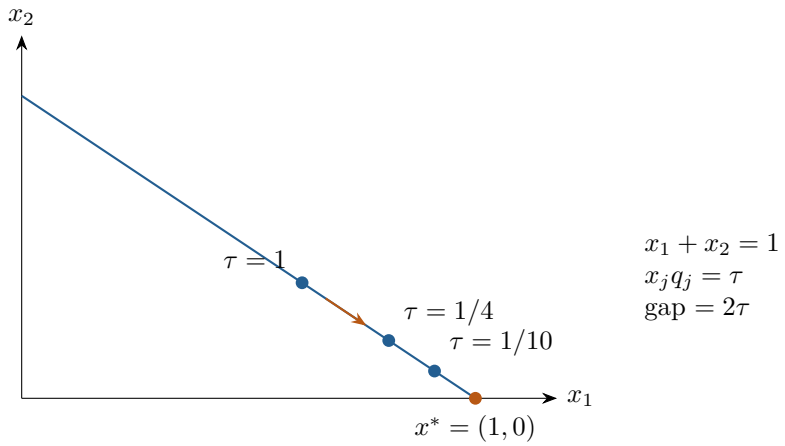


Figure 12: Computed central-path locations from the explicit formula, rather than solver iterates. The relative interior here is a line segment; decreasing the barrier parameter approaches its optimal endpoint.

Strict primal feasibility alone does not guarantee a barrier minimizer. For example, adding a variable fixed only by $x_3 \geq 0$ with zero cost makes $-\tau \log x_3$ unbounded below as $x_3 \to \infty$. Rows forcing a variable to zero also leave this simple barrier with no domain. Such cases require reformulation or a more general initialization strategy. Primal–dual methods solve perturbed KKT equations using Newton steps, with step lengths controlling positivity and residuals. Their convergence and complexity require additional analysis; the one-dimensional formula is an illustration, not a general implementation.

14 MATLAB checks and a study checklist

The matching base MATLAB script is `matlab/exercises/lecture_05_checks.m`. From the repository root:

```
run('matlab/exercises/lecture_05_checks.m')
```

The basic calculations use linear solves:

```
AB = A(:, B);
T = AB \ A;
basic = AB \ b;
y = AB' \ c(B);
q = A' * y - c;
value = c(B)' * basic;
```

The script checks the same exact inputs as the hand solutions with absolute tolerance 10^{-10}. It verifies selected pivots and optimality certificates, and enumerates all nonsingular bases in these small examples to compare finite optima. Enumeration is a cross-check for the supplied models, not a general solver or a test for every kind of unboundedness. No Optimization Toolbox is required.

Hand result	Expected value
Main example, second RHS change	$-50/3 \le t \le 22$
Main example, decision-cost ranges	$c_1' \ge -3/2,\ c_2' \ge 2,\ c_3' \le 2/5$
Small model, new objective $(3,4,2)$	$x = (8/3, 10/3, 0),\ z = 64/3$
Small model, RHS $(3,4)$	$x = (3,0,0),\ z = 6$
Small model, added column	$x_1 = 16,\ x_6 = 10,\ z = 42$
Small model, added inequality	$x = (10/3, 0, 8/3),\ z = 28/3$
Small model, delete x_1	$(x_2, x_3) = (0,6),\ z = 6$
Production, remove first row	$x = (4,4),\ z = 20$
Mixed-sign KKT certificate	$\hat{x} = (0, 17/4, 0, 23/4),\ z = 37/2$

14.1 Common mistakes

Mistake	Correction
Confusing a new coefficient with its change	Use $b_r' = b_r + t$ or $c_k' = c_k + t$
Claiming unchanged basis means unchanged point	RHS changes move the basic vector
Combining individual intervals independently	Test the complete vectors together
Reading every slack or artificial as $+e_r$	Check the original column and its sign
Appending a raw constraint to a final tableau	Eliminate the old basic columns first
Calling every basic artificial infeasible	Its optimal value must be strictly positive
Treating a slack row as globally redundant	Deleting it can enlarge the feasible set
Using positive multipliers on a $\ge$ normal	Flip the row to $\le$ first
Ignoring bound normals in stationarity	Include $-x_k \le 0$
Claiming convexity alone guarantees KKT necessity	Check a constraint qualification

14.2 Questions to test understanding

1. In (5), what happens when $b_2' = 40$?
   Answer: the old basic slack is -3, so that basis fails. The new optimum is $(31/5, 58/5, 0)$, value $143/5$, second slack 6.

2. What changes when $c_1' = -2$?
   Answer: $t = -1 < -1/2$. The old point is feasible, but its $q_4 = -1/5$. Entering x_4 has a positive step and improves the new objective.

3. For (11), change both b to $(3, 4)$ and costs to $(4, -1, 1)$. Why is the value not $12 + (-6) + 12$?
   Answer: the basis passes both tests; its new point is $(3, 0, 0)$ and value 12. The separate first-order changes miss the cross term -6.

4. What if a new variable has $a = 0$, $\gamma = 1$?
   Answer: $q_w = -1$, $u = 0$. The original point plus $w = \theta$ is a feasible improving ray, so the extended maximum is unbounded.

5. Can a tight constraint be deleted without changing the optimum?
   Answer: yes. For example, duplicate $x_1 \leq 1$ in $\max x_1$, $x_1 \geq 0$; deleting either copy leaves the other bound.

6. Why do positive μ_1, μ_3 prove uniqueness in the KKT example?
   Answer: any optimizer must satisfy $x_1 = x_3 = 0$ by zero dual gap. The positive third-row multiplier forces that row tight; with the equality, these determine x_2, x_4 uniquely.

15 Sources and editorial remarks

The primary source is E. Pawelec, Wydział MiNI PW, `s4_w5.pdf`, in the external historical lecture folder. It has 10 PDF pages containing printed slides 1–38, with the last slide a closing graphic. The metadata creation date is 2019-11-03; this does not establish the delivery date. The folder is labelled 2020. Alignment with the current lecturer, syllabus, exercises, and labs is unconfirmed.

Historical coverage	Corresponding development here
Slides 1–8: standard form, tools, analysis types	Sections 1 and 5
Slides 9–14: one-parameter ranges	Sections 2–3; Polish exercise prompts retained
Slides 15–19: vector changes	Section 4; unfinished objective calculation completed
Slides 20–22: new column	Section 6
Slides 23–27: new row	Section 7; unfinished dual pivot completed
Slide 28: remove a variable	Section 8; feasibility phase made explicit
Slides 29–33: remove a row	Section 9; useful production figures redrawn in TikZ
Slides 34–37: Lagrange multipliers	Section 10; candidate question retained and fully certified

Added material includes the basic-value and indicator plots, complete piecewise value function with proofs, data-space regions, simultaneous-change counterexample and cross term, structural certificates, feasible-direction and normal-cone diagrams, uniqueness proof, study questions, and MATLAB checks.

Qualifications correct overly broad shortcuts: simultaneous changes are not inherently unpredictable; a failed basis indicator test is not always a failed point-optimality test; positive artificial value, not basis membership, is an infeasibility certificate; general nonlinear convex KKT necessity needs regularity. The structural example with a negative new-column component is treated as an algebraic LP update, without inventing a physical production interpretation.

Two official supplementary references were consulted to cross-check terminology and hypotheses, with maximization signs adapted consistently:

- MIT OCW 6.251J/15.081J, Fall 2009, Lecture 12: Sensitivity analysis, PDF pp. 2–6: basis tests, one-parameter updates, vector interpretation, new columns and rows. See Sections 2–7.
- L. Vandenberghe, UCLA ECE236B, Winter 2026, Chapter 5: Duality, sections on Slater's qualification, optimality conditions, and sensitivity. See Sections 10–11.

These are supplements, not evidence of current course coverage. External download locations and status are recorded in the paired source inventory.

Sections 12–13 are added bridges to nonlinear and interior-point optimization, with unknown current-course alignment. The KKT qualifications and barrier terminology were cross-checked against Boyd and Vandenberghe, *Convex Optimization*, Sections 5.2.3, 5.5.3 and 11.2. The cubic, equality-constrained curvature and two-variable central-path examples are original derivations; Figure 12 uses their explicit formula. They are not covered by the existing MATLAB companion.

Verification: the hand tableaus, endpoints, structural updates, and KKT identity were checked independently with exact rational arithmetic. All Lecture 5 MATLAB assertions passed in the user's normal terminal on 2026-10-04, with reported version 26.2.0.3386108 (R2026b). The agent's launcher still exited silently; this is user-supplied execution evidence. The paired verification log records the exact command and output.